

Simulating AR(1) and MA(1) Processes in R

A Reproducible Introduction to Linear Time-Series Dependence

Kalle Saarelma

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Abstract

Autoregressive and moving-average processes provide the elementary building blocks of linear time-series analysis. This article develops a compact, reproducible implementation of first-order autoregressive, AR(1), and first-order moving-average, MA(1), processes in base R. The implementation separates simulation, visualization, and validation into reusable components and records all parameter values and random seeds. Simulated trajectories and sample autocorrelation functions illustrate the central theoretical distinction between the models: the AR(1) autocorrelation decays geometrically, whereas the MA(1) autocorrelation cuts off after lag one. The project demonstrates how a small amount of well-structured code can connect formal time-series theory with transparent computational experiments.

Keywords: autoregression, moving average, autocorrelation, stationarity, reproducible research, R

1 Introduction

Time-series observations are ordered, and their temporal order is often informative. Economic indicators, sensor measurements, financial returns, and environmental records may all exhibit dependence between present and past values. Linear stochastic processes offer an interpretable framework for representing such dependence. In particular, autoregressive and moving-average models form the basis of the ARMA and ARIMA families used throughout applied forecasting and econometrics [1, 2].

This article studies the simplest nontrivial members of these families: the AR(1) and MA(1) models. Although elementary, they display qualitatively different dependence structures. An AR(1) process propagates earlier values forward through a recursive equation. An MA(1) process, by contrast, depends only on the current and immediately preceding shocks. Simulation makes this contrast visible and provides a useful check on analytical results.

The accompanying R project has three aims. First, it implements both models without external package dependencies. Second, it produces repeatable figures and data by fixing random seeds. Third, it includes lightweight checks that verify output dimensions and deterministic reproduction. The complete source is available at <https://github.com/kjsaarel/time-series-processes>.

2 Model definitions and properties

2.1 The AR(1) process

Let $\{\varepsilon_t\}$ be independent and identically distributed innovations with mean zero and variance σ_ε^2 . An AR(1) process satisfies

$$y_t = \phi y_{t-1} + \varepsilon_t, \tag{1}$$

where ϕ controls persistence. If $|\phi| < 1$, the process has a covariance-stationary solution with

$$\text{Var}(y_t) = \frac{\sigma_\varepsilon^2}{1 - \phi^2}. \quad (2)$$

Its theoretical autocorrelation function (ACF) is

$$\rho(k) = \phi^k, \quad k = 0, 1, 2, \dots \quad (3)$$

Positive values of ϕ produce persistence with a monotone geometric decay. Negative values produce an alternating ACF. When $\phi = 1$, Equation (1) becomes a random walk and is no longer covariance stationary.

2.2 The MA(1) process

An MA(1) process is defined by

$$y_t = \varepsilon_t + \theta \varepsilon_{t-1}, \quad (4)$$

where θ determines the contribution of the previous innovation. The variance and ACF are

$$\text{Var}(y_t) = (1 + \theta^2)\sigma_\varepsilon^2, \quad (5)$$

$$\rho(1) = \frac{\theta}{1 + \theta^2}, \quad (6)$$

$$\rho(k) = 0, \quad k \geq 2. \quad (7)$$

Thus, the MA(1) process is stationary for every finite θ . The commonly imposed condition $|\theta| < 1$ concerns invertibility: it ensures a unique, convergent autoregressive representation of the model [3].

3 Computational design

The project separates the computational workflow into three layers. The file `R/simulate_processes.R` contains functions for simulation and plotting. The script `scripts/run_demo.R` fixes the experimental parameters, generates both series, and writes figures and comma-separated data files. Finally, `tests/test_simulations.R` checks vector lengths, initial conditions, and reproducibility under fixed seeds.

Table 1 summarizes the baseline experiments. Innovations are drawn from a standard normal distribution, so $\sigma_\varepsilon^2 = 1$.

Table 1: Simulation parameters.

Model	Observations	Parameter	Seed	Initial value
AR(1)	150	$\phi = 0.9$	12	$y_1 = 0$
MA(1)	100	$\theta = 0.5$	123	Not applicable

The AR(1) implementation uses an explicit recursion. This choice mirrors the mathematical definition and makes the role of the initial condition transparent. The MA(1) implementation draws $n + 1$ innovations, including the pre-sample shock ε_0 , and then evaluates Equation (4) vectorially. Both simulation functions return the realized series, innovations, and model parameter in a structured list.

Reproducibility requires more than setting a seed in an interactive session. The parameters, seeds, output locations, and execution commands must also be recorded. The complete experiment can be regenerated from the repository root with

```
Rscript scripts/run_demo.R
Rscript tests/test_simulations.R
```

No contributed R packages are required.

4 Simulation results

Figure 1 shows the AR(1) realization and its sample ACF. With $\phi = 0.9$, shocks have long-lived effects. The simulated trajectory therefore exhibits extended excursions above and below zero rather than rapidly oscillating around the mean. The sample ACF remains strongly positive at the first several lags and declines gradually, in agreement with the geometric pattern predicted by Equation (3). Sampling variation prevents the empirical correlations from lying exactly on the theoretical sequence.

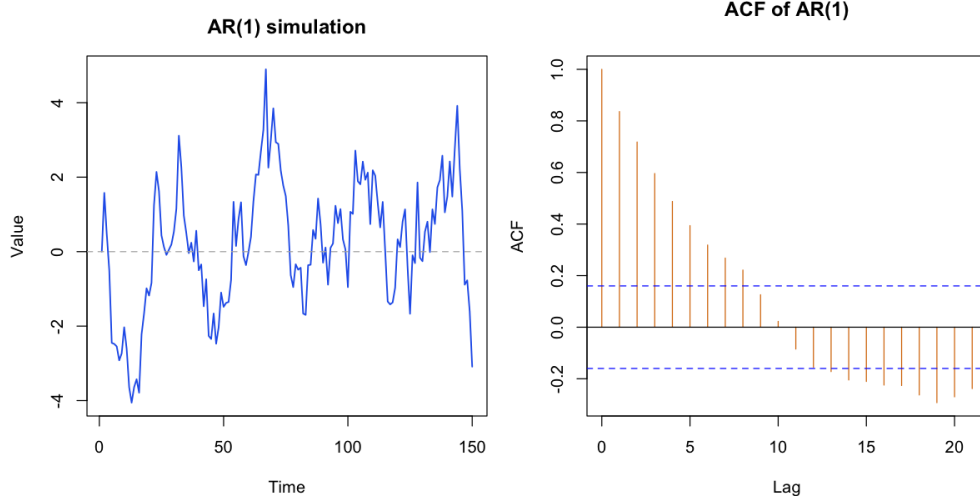


Figure 1: A simulated AR(1) process with $\phi = 0.9$ (left) and its sample autocorrelation function (right).

Figure 2 displays the MA(1) experiment. Compared with the persistent AR(1) trajectory, the MA(1) series reverses direction more rapidly because an innovation affects only two adjacent observations. For $\theta = 0.5$, the theoretical first-lag correlation is

$$\rho(1) = \frac{0.5}{1 + 0.5^2} = 0.4.$$

The sample ACF shows a positive first-lag association and correlations near zero thereafter. This abrupt cutoff is the diagnostic theoretical signature of an MA(1) process.

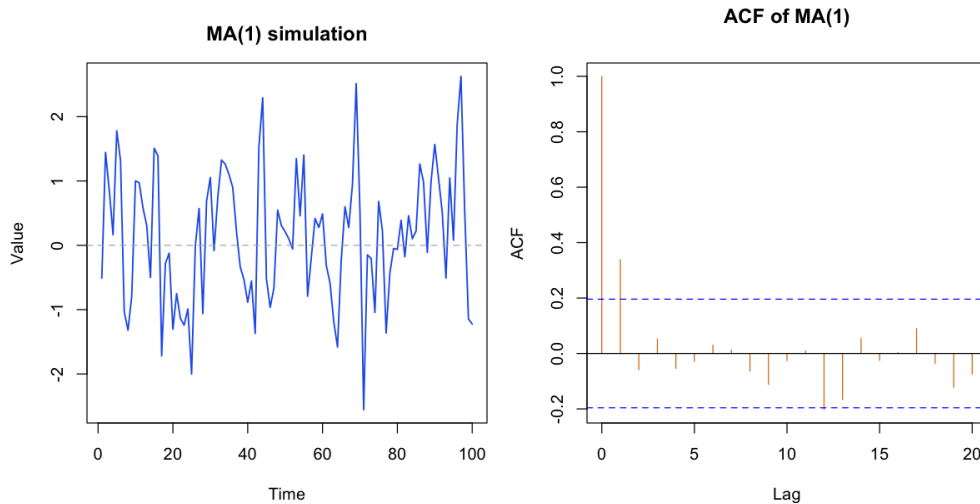


Figure 2: A simulated MA(1) process with $\theta = 0.5$ (left) and its sample autocorrelation function (right).

5 Discussion

The experiment illustrates how model structure determines serial dependence. In an AR(1) model, a shock influences y_t directly, y_{t+1} through ϕ , y_{t+2} through ϕ^2 , and so forth. The effect diminishes but does not terminate after a fixed lag. In an MA(1) model, the same shock enters only two consecutive observations. Consequently, the population ACF cuts off exactly after lag one.

These observations also motivate a classical identification heuristic. An autoregressive model tends to have a gradually decaying ACF and a partial autocorrelation function that cuts off. A moving-average model tends to display the reverse pattern [4]. In finite samples, however, the distinction is imperfect. Sample autocorrelations are noisy, and a highly persistent AR process can resemble a nonstationary series. Model selection should therefore combine graphical diagnostics with estimation, information criteria, residual analysis, and substantive knowledge.

The present exercise is deliberately limited. It assumes Gaussian, homoskedastic innovations; it does not estimate parameters from observed data; and it considers only one realization per parameter setting. A natural extension would compare empirical and theoretical ACFs across repeated simulations. Further extensions could examine negative parameters, near-unit-root behavior, forecast uncertainty, or mixed ARMA processes.

6 Conclusion

AR(1) and MA(1) processes encode two fundamentally different mechanisms of time dependence. Autoregression transmits past values recursively and produces geometrically decaying autocorrelation. Moving-average dependence transmits recent innovations over a finite horizon and produces an ACF cutoff. The reproducible R project developed here makes these properties visible while maintaining a clear separation among simulation, output generation, and validation. This combination of mathematical derivation and transparent code provides a practical foundation for more advanced time-series modeling.

References

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